

Logical Database Design

What is Logical Database Design?

- Logical database design is the process of designing the logical structure of a database.
- It converts the conceptual design (ER model) into tables and relationships.
- It focuses on data organization, not physical storage.

Definition

Logical Database Design:

The process of converting the ER model into relational tables with proper relationships and constraints.

Objectives of Logical Database Design

1. Organize data efficiently
2. Reduce redundancy
3. Maintain consistency
4. Establish relationships
5. Improve database performance

Features of Logical Database Design

6. Based on conceptual design
7. Uses tables and relations
8. Independent of physical storage
9. Defines keys and constraints
10. Supports normalization

Components of Logical Database Design

1. Tables (Relations)

Definition

- Data is stored in tables.

Example

Student_ID Name Course

101 Ravi CSE

Important Points

- 11.Rows are called tuples
- 12.Columns are called attributes
- 13.Each table has unique name

2. Attributes

Definition

- Columns in a table.

Example

- Student_ID
- Name
- Age

Important Points

- 14.Describe table properties
- 15.Each attribute has data type

3. Keys

Primary Key

Definition

- Uniquely identifies each record.

Example

- Student_ID

Important Points

16. Cannot contain NULL

17. Must be unique

Foreign Key

Definition

- Connects one table with another table.

Example

- Dept_ID in Student table
-

Important Points

18. Maintains relationships

19. References primary key

4. Relationships

Definition

- Association between tables.

Types

20. One-to-One

21. One-to-Many

22. Many-to-Many

Steps in Logical Database Design

Step 1: Create Tables

- Convert entities into tables.

Example

Entity: Student

Table: Student

Step 2: Define Attributes

- Add columns to tables.

Example

Student Table:

- Student_ID
- Name
- Course

Step 3: Identify Primary Keys

- Select unique identifiers.

Example

- Student_ID
-

Step 4: Define Relationships

- Connect tables using foreign keys.

Example

- Dept_ID references Department table

Step 5: Apply Constraints

- Add rules for data validity.

Example

- NOT NULL
- UNIQUE

Step 6: Normalize Tables

- Remove redundancy and anomalies.

Normalization in Logical Design

1. First Normal Form (1NF)

23. Removes repeating groups

2. Second Normal Form (2NF)

24. Removes partial dependency

3. Third Normal Form (3NF)

25. Removes transitive dependency

Example of Logical Database Design

Student Table

Student_ID	Name	Dept_ID
101	Ravi	D1

Department Table

Dept_ID	Dept_Name
D1	CSE

Relationship

- Dept_ID in Student table references Department table.

Integrity Constraints in Logical Design

Entity Integrity

26. Primary key cannot be NULL

Referential Integrity

27. Foreign key must match primary key

Domain Integrity

28. Attribute values must be valid

Advantages of Logical Database Design

29. Reduces data redundancy
30. Improves consistency
31. Simplifies database maintenance
32. Improves data integrity
33. Supports efficient querying
34. Makes relationships clear
35. Easier implementation
36. Better scalability
37. Improves performance
38. Helps in normalization

Additional Important Points

39. Logical design follows conceptual design.
40. ER model is converted into relations.

41. Tables represent entities.
42. Attributes represent properties.
43. Relationships use foreign keys.
44. Normalization improves efficiency.
45. Constraints maintain consistency.
46. Logical design is DBMS independent.
47. Logical schema defines table structure.
48. Proper logical design reduces anomalies.
49. Logical design supports data sharing.
50. Logical database design is important before physical implementation.

Difference Between Conceptual and Logical Design

Conceptual Design

Uses ER model

High-level view

Focuses on entities

No implementation details

Logical Design

Uses tables

Detailed structure

Focuses on relations

Defines keys and constraints

Real-Time Example

Library Management System

Tables

- Book
- Member
- Issue

Relationships

- Member borrows Book

Keys

- Book_ID
- Member_ID

Short Definitions for Exams

Logical Database Design

The process of converting conceptual database design into tables and relationships.

Logical Schema

A structure that defines tables, attributes, keys, and relationships.

shyam-sudheer1602